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Common institutional ownership and cost of equity: evidence from publicly listed companies in China

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ABSTRACT
Using data from 2682 publicly listed companies in China between 2012 and 2022, the study employs panel data regression to investigate the impact of common institutional ownership (CIO) on the cost of equity (COE). The findings indicate that CIO significantly reduces COE, with this reduction attributed to two main factors: the exit threat of common institutional investors (CIIs) and the synergistic governance effect. The results remain robust across alternative measures of COE using the PEG and MPEG models proposed by Easton (2004), as well as endogeneity tests utilizing Two Stage Least Squares. Moreover, the study identifies several factors that influence the strength of the CIO-COE relationship, including industry competitiveness, corporate governance quality, and financing constraints. The findings highlight the importance of leveraging the advantages of CIO in corporate supervision and governance, while also emphasizing the need for effective regulation to prevent collusive behavior and promote market efficiency.

IMPACT STATEMENT
The implications of the study contribute to a better understanding of the role of CIO in shaping the COE and have implications for policymakers, regulators, and market participants seeking to enhance corporate governance practices and reduce the COE.

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Common institutional ownership (CIO); cost of equity (COE); common institutional investors (CIIs); industry competition; corporate governance; financial constraint

SUBJECTS
Finance; Business, Management and Accounting; Economics

1. Introduction

Common institutional ownership (CIO) refers to the situation where the same institutional investors simultaneously hold shares in multiple companies in the same industry. According to Lewellen and Lowry (2021), only 17% of the companies in the S&P 500 Index had CIO in 1999. By the end of 2015, this proportion had increased to 81%.

One vein of the literature suggests that CIO facilitates synergistic governance among publicly listed companies (PLCs) within an industry, provides convenience for information exchange and resource sharing, promotes positive cooperation, and reduces unnecessary costs of destructive competition (Koch et al., 2021). For example, Edmans et al. (2019) argue that under common ownership, management has a higher motivation for active corporate governance because common institutional investors (CIIs) can use their information advantage on the entire industry to sell stocks, transmitting negative signals to the capital market and increasing pressure on managers. Park et al. (2019) confirm that CIO helps reduce malicious competition within an industry, improve market liquidity, decrease proprietary information disclosure costs for PLCs, and enhance corporate governance. Gao et al. (2019) argues that the portfolio maximization objective will prompt CIIs to coordinate actions among PLCs in the same industry within their portfolio, thereby enhancing corporate governance and supervision. Ramalingegowda et al. (2021) finds that higher CIO is related to less earnings management. Dennis et al. (2022) demonstrate that there is an increase in voluntary information disclosure among firms as the level of CIO rises, indicating that CIO reduces the costs of information disclosure and improves industry coordination mechanisms

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among PLCs. Wang et al. (2023), Gao et al. (2024) and Chen et al. (2024) show that CIIs, with advantages in information, experience, and resources, can effectively inhibit corporate misconduct, earnings management and stock price crash risk, respectively.

Another vein of literature has raised concerns about potential monopolistic practices and collusive fraud in CIO. For example, Azar et al. (2018) ests that CIIs are more inclined to pursue short-term gains without considering long-term development, which may lead to collusion with major shareholders to encroach on the interests of minority shareholders. Lewellen and Lowry (2021) evaluates various empirical approaches to identify the effects of CIO and finds little robust evidence that CIO positively affects firm profitability, investment, mergers, or strategic alliances. Pawliczek et al. (2022) believes that CIIs aim to maximize profits and have a strong motivation to collude with companies. Corporate collusion not only reduces the positive effects of competition among companies but also lowers investment sensitivity, ultimately causing companies to miss investment opportunities and reduce investment efficiency. Cheng et al. (2022) finds that CIO is negatively associated with the level of corporate social responsibility, which supports an anti-competitive view. Additionally, the anti-competitive effect is more pronounced for mature firms, and for firms in industries with lower labor intensity and lower customer sensitivity. Kini et al. (2024) finds that common ownership can inhibit product market competition and dynamism, especially in industries more prone to quasi-monopoly outcomes in product spaces. Antón et al. (2023) finds that managerial incentives are less performance sensitive in firms with more common ownership.

Given the lack of consensus in the academic literature, it is evident that further research is needed to gain a comprehensive understanding of the implications of CIO. Conducting studies that employ diverse methodologies and data sources can provide valuable insights into the potential effects of CIO on various aspects of corporate governance, performance, and strategic decision-making. However, most studies focus on developed capital markets, while only a few (e.g., Bai et al., 2023; Li & Liu, 2023; Wicaksono et al., 2024; Xu et al., 2023) explore the effects of CIO in emerging markets. Therefore, the objective of this article is to examine whether and to what extent CIO influences the COE, using a dataset comprising 2682 Chinese PLCs during the period from 2012 to 2022.

The novelties and marginal contributions of this article are primarily manifested in the following three aspects: Firstly, while prior research on CIO has primarily focused on its effects on corporate information disclosure (Park et al., 2019; Pawliczek et al., 2022), innovation efficiency (Gao et al., 2019; Li et al., 2023), mergers and acquisitions (Lewellen & Lowry, 2021), earnings management (Gao et al., 2024; Ramalingegowda et al., 2021), financial misconduct (Wang et al., 2023) and stock price crash risk (Chen et al., 2024) this article represents one of the first efforts to investigate the influence of CIO on COE. By exploring this relationship, the article enriches the literature on the governance role of CIO and extends our understanding of how CIO may affect firm valuation.

Secondly, while previous studies have examined the impact of institutional investors in general on COE (Edmans et al., 2019; Gillan & Starks, 2000; Gompers et al., 2003; He & Huang, 2017), this article specifically focuses on CIIs and investigates their influence on COE through the lenses of 'exit threat' and 'collaborative governance' within the framework of CIO. This focused examination allows for a deeper exploration of the mechanisms through which CIIs affect COE, thereby extending existing research on the determinants of COE.

Thirdly, this article goes beyond merely examining the direct relationship between CIO and COE by exploring the moderating roles of industry competition, corporate governance, and financial constraints. By considering these contextual factors, the study aims to illuminate the complex interplay between CIO and COE. Specifically, it investigates whether levels of industry competition, the quality of corporate governance practices, and financing constraints influence the strength or direction of the relationship between CIO and COE. This analysis provides a nuanced understanding of the underlying mechanisms through which CIO affects COE and offers empirical evidence that can inform the formulation of relevant policies by regulatory authorities.

2. Theoretical analysis and hypothesis development

Under traditional asset pricing theories, CIO would have to be related to a firm's systematic risk to be non-diversifiable and therefore priced into the firm's cost of equity capital. According to existing studies,

CIO can reduce the COE through three different channels: Price informativeness, product market risks and synergistic governance effects. Specifically, CIO can enhance the transparency and informativeness of a firm's financial data. This improved disclosure quality allows investors to better assess the firm's expected value and future cash flows, leading to a reduction in uncertainty and, consequently, a lower COE (Boone & White, 2015; Edmans et al., 2019; Wu et al., 2023). In addition, strategic alliances and incentives to avoid competition in the product market allow firms to access larger markets, benefit from economies of scale, and internalize negative externalities. This reduces a firm's product market risk, which is partially non-diversifiable, ultimately lowering the COE (Bloomfield et al., 2024; Koch et al., 2021; Park et al., 2019). Furthermore, CIIs have both the incentives and the ability to influence their portfolio firms' cost of capital through corporate governance and disclosure changes. By promoting better corporate governance practices and improving disclosure quality, CIIs can enhance investor confidence and reduce the uncertainty surrounding future cash flows. This, in turn, reduces the portfolio firms' COE (Kang et al., 2018; Ni & Yin, 2023).

Firstly, in terms of price informativeness, CIIs typically hold a diversified portfolio consisting of relatively good and relatively poor PLCs. When faced with liquidity risk or other crises, CIIs can flexibly decide which assets to sell. In cases where CIIs hold more than 5% of the shares, their divestment can have a detrimental impact on the underlying PLCs. The first assets to be sold are often the relatively underperforming ones, which signal negative information of the companies to the market. This process leads to a survival-of-the-fittest dynamic, exerting pressure on PLCs from various aspects and compelling them to strengthen their governance practices through the mechanism of 'exit threat' (Boone & White, 2015; Edmans et al., 2019; Koch et al., 2021). Such a threat can lead to enhanced disclosure quality, greater long-term shareholdings, and higher risk-absorption capacity (Khoo et al., 2024). In the Chinese capital market, where herd behavior is significant, price information signals are more easily captured by investors, making the informational content of price reactions deeper under the CIO framework. This, in turn, compels managers to improve information disclosure and increase corporate transparency, which can lead to lower COE (Wu et al., 2023).

Secondly, in terms of product market risks, CIO creates strategic alliances and incentives that enable companies to collaborate rather than compete, yielding multiple economic and operational benefits (Park et al., 2019). For example, firms can leverage their partners' market presence, distribution networks, and customer bases. The increased market access can lead to higher sales volumes and revenue streams, enhancing the firm's financial stability and reducing the variability in cash flows. This stability is attractive to investors, as it signifies a lower risk profile (Gao et al., 2024). In addition, when companies collaborate, they can pool resources, share R&D costs, and optimize production processes. The resulting efficiencies from larger-scale operations can lead to cost reductions in manufacturing, procurement, and distribution (Bai et al., 2023; Li & Liu, 2023). These cost savings improve profit margins and provide a competitive edge in pricing. Enhanced profitability and cost efficiency make the firm more resilient to market fluctuations and economic downturns, further lowering the perceived risk by investors (Bloomfield et al., 2024).

Thirdly, from the perspective of synergistic governance effects under the CIO framework, CIIs with stakes in multiple companies within the same industry have a greater advantage in terms of information acquisition, industry experience, and corporate influence. Specifically, CIIs can accumulate in-depth industry-specific knowledge, information, and experience, making it easier for them to identify inaccurate reports and exercise interventions. The governance experience obtained can reduce monitoring costs and information uncertainties, enhancing the effectiveness of CIIs' supervision (Kang et al., 2018). Moreover, CIIs' involvement reduces the competitiveness among companies in the same industry, thereby lowering the possibility and costs of destructive competition (Gilje et al., 2020). Overall, the CIO framework helps companies to coordinate industry product market strategies, resolve disputes, reduce destructive competition, and promote positive cooperation (Kini et al., 2024; Liu & He, 2024). This may lead to a reduction in stock price volatility and investors' risk expectations towards the PLC, and ultimately help lower its COE.

Based on the above analysis, this study proposes the following hypothesis:

[H1]: *Firms with common institutional ownership have lower cost of equity capital.*

For companies operating in highly competitive industries, the intensity of competition raises concerns about the survival of businesses. Moreover, stakeholders are wary of contractual loopholes that result in low levels of inter-firm cooperation and inefficiencies. Specifically, intense industry competition exposes companies to increased risks and unethical practices. In order to avoid being eliminated in such fierce competition, companies in highly competitive environments are more motivated to engage in proactive corporate governance to enhance their value and reduce their COE compared to companies in less competitive environments (He & Huang, 2017).

Additionally, driven by the goal of maximizing individual firm value, the higher the competition within an industry, the more likely businesses impose externalities on each other, leading to inefficiencies. Considering the exit mechanism of CIO, companies facing higher levels of industry competition release more sensitive signals to the market through the CIO framework. This increased sensitivity creates stronger incentives for better corporate governance, which ultimately contributes to reducing a company's COE.

Moreover, the structure of the product market affects a company's cash flow risk and equilibrium return (Li et al., 2023). Through common ownership, competition and inefficiency among companies in the investment portfolio can be suppressed, and common owners can reduce a company's product market risk (Bloomfield et al., 2024). Therefore, as market competition intensifies, the negative correlation between CIO and a company's COE becomes stronger.

Based on these considerations, this study proposes the following hypothesis:

[H2]: *The higher the level of industry competition, the more significant the role of common institutional ownership in reducing a company's cost of equity capital.*

Good corporate governance mechanisms have several benefits. On one hand, they help alleviate agency problems among management and provide effective supervision of PLCs (Gillan & Starks, 2000; Schäuble, 2019). On the other hand, they contribute to improving information asymmetry issues and enhancing the quality of information disclosure by PLCs (Hartzell & Starks, 2003). As hypothesized earlier, from the perspective of the synergistic governance effect of CIO, whether through active participation in corporate governance supervision or by influencing executive behavior, CIIs can achieve the goal of conveying positive signals through proactive corporate governance. This, in turn, reduces investors' risk expectations and lowers a company's COE.

From the perspective of information asymmetry, CIIs, driven by the maximization of portfolio value, often tend to integrate and coordinate industry experience, information resources, economies of scale, and other factors within their portfolios, resulting in integration effects (He et al., 2019). This is beneficial for reducing cutthroat competition. In other words, when a company has a high level of corporate governance, it not only enables CIIs to exert synergistic governance effects but also facilitates their integration of resources and information gathering, leading to economies of scale (Dennis et al., 2022). This, in turn, reduces stock price volatility and investors' risk expectations toward the PLC, ultimately lowering the company's COE.

Based on these considerations, this study proposes the following hypothesis:

[H3]: *The higher the level of corporate governance, the more significant the role of common institutional ownership in reducing a company's cost of equity capital.*

The level of financial constraints on a PLC directly affects its financing costs, which in turn impact its investment capabilities, production scale, and efficiency. Higher financial constraints lead to increased financing costs, hindering the company's ability to invest and affecting its overall operations. Conversely, lower financial constraints enable PLCs to secure external financing more easily, facilitating timely investments in profitable projects and improving business performance (Parker & van Praag, 2006).

For PLCs facing significant financial constraints, CIIs can play a crucial role in fostering market cooperation and facilitating strategic acquisitions among their portfolio companies. By mitigating inefficiencies and reducing cutthroat competition, CIIs enhance innovation, productivity, and profitability, thereby alleviating the financial pressures faced by these firms (Gao et al., 2019). Research by Du and Zhang (2023) further underscores the positive role of CIOs in reducing uncertainty in investment outcomes and

increasing financial flexibility. The presence of a CIO becomes particularly valuable as financial constraints on a PLC intensify, contributing significantly to reducing the company's COE.

CIIs not only reduce competition within the industry but also facilitate the flow of resources and information among companies, leading to improved profitability and production efficiency (Wu et al., 2023). This collaboration helps companies overcome high financial constraints and harnesses the synergistic governance effects of the CIO, ultimately lowering equity costs. Consequently, the more financially constrained a PLC is, the more pronounced the CIO's negative impact on the company's COE.

Based on these considerations, this study proposes the following hypothesis:

[H4]: *The higher the level of financial constraint, the more significant the role of common institutional ownership in reducing a company's cost of equity capital.*

3. Data and methodology

This study selects 2682 non-financial listed companies in the A-share market of China from 2012 to 2022 as the research sample. To ensure the validity of the sample, we follow the convention and delete all financial and insurance companies, firms tagged as Special Treatment (ST) due to consecutive losses in over two years, and firms with missing data on key variables. The indicator of CIO is manually collected and compiled based on quarterly financial reports available from the China Securities Market and Accounting Research (CSMAR) database[1]. Other variables are obtained from both the CSMAR and Wind database. To mitigate the impact of outliers on the regression, a winsorization process was employed on all continuous variables within the regression model, trimming the upper and lower 1% of the dataset. Ultimately, a total of 35,635 observations was obtained.

To test the hypothesis, this paper specifies the following panel data regression model:

$$PEG_{i,t} = \beta_0 + \beta_1 CIO_{i,t} + \delta \mathbf{X}_{it} + \varphi_i + \lambda_t + \varepsilon_{it} \tag{1}$$

where $PEG_{i,t}$ represents the cost of equity capital of company i during year t, calculated based on the PEG model; $CIO_{i,t}$ Represents the status of common institutional ownership in the listed company. The parameter β_1 captures the impact of the CIO on COE. The matrix $\mathbf{X}_{it}$ Is a vector of control variables that prior studies have shown to explain the cross-section of expected equity returns including firm size (Fama & French, 1992, 2015), leverage ratio (Fama & French, 1992), profitability (Fama & French, 2015), market-to-book ratio (Fama & French, 1992, 2015), growth opportunities (Hou et al., 2015), and corporate governance indicators such as board size, the proportion of independent directors, and institutional investor ownership (Core et al., 2006; Ghosh et al., 2022; Gompers et al., 2003; Huo et al., 2021). The term φ_i denotes firm fixed effects, λ_t denotes year-fixed effects, and the error term ε_{it} accounts for random disturbances. To address potential autocorrelation at the firm level, cluster-robust standard errors are computed in all regressions. Hypothesis requires $\hat{\beta}_1 < 0$.

This study utilizes the PEG model proposed by Easton (2004) to calibrate COE. Specifically, the PEG model calculates the COE based on the relationship between a company's price-earnings ratio (P/E ratio) and its expected earnings growth rate. It assumes that the growth rate of earnings is constant, and the formula for the COE is as follows: $PEG_{i,t} = \sqrt{\frac{(eps_{i,t+1} - eps_{i,t})}{P_{i,t}}}$ where $eps_{i,t+2}$ and $eps_{i,t+1}$ are earnings per share for company i during year $t+2$ and $t+1$, respectively, and $P_{i,t}$ is its share price at the end of year t.

Drawing on existing literature (Chen et al., 2024; He & Huang, 2017; Koch et al., 2021), this study constructs two indicators to measure CIO of listed companies. The first indicator is a CIO dummy variable (CIO_1). If a PLC has CIIs holding shares in any single quarter of the year, CIO_1 takes the value 1; otherwise, it takes value 0. CIIs refer to institutional investors holding at least a 5% stake in two or more companies in the same industry. The second indicator is the degree of connectivity of common ownership (CIO_2), which is the number of CIIs holding shares in a PLC.

It is worth noting that CIO_1 and CIO_2 are constructed based on quarterly data. If a PLC has CIIs in any quarter of a given year, it is considered to have been held by CIIs in that year. The average of the quarterly indicator is taken as the corresponding annual indicator data.

To examine the moderating role of industry competition, this paper augments model (1) by including the interaction of the level of industry competition (HHI_{it}) with the CIO variable and creates the following model:

$$PEG_{i,t} = \beta_0 + \beta_1 CIO_{i,t} + \beta_2 CIO_{i,t} \times HHI_{i,t} + \delta \mathbf{X}_{it} + \varphi_i + \lambda_t + \varepsilon_{it} \quad (2)$$

where HHI_{it} Is the Herfindahl-Hirschman Index calculated by summing the squares of the market shares of all firms in an industry. A higher value of this index indicates a higher level of industry concentration and less degree of competition. Hypothesis requires $\hat{\beta}_2 > 0$.

To examine the moderating role of corporate governance (CG_{it}), this paper augments model (1) by including the interaction of CG_{it} with the CIO variable and creates the following model:

$$PEG_{i,t} = \beta_0 + \beta_1 CIO_{i,t} + \beta_2 CIO_{i,t} \times CG_{i,t} + \delta \mathbf{X}_{it} + \varphi_i + \lambda_t + \varepsilon_{it} \quad (3)$$

where CG_{it} represents the level of corporate governance, which is constructed by selecting eight variables related to different aspects of corporate governance using principal component analysis, namely, the separation of the chairman and CEO positions, the proportion of independent directors, the proportion of board ownership, the proportion of ownership by the largest shareholder, the proportion of executive ownership, the size of the board of directors, the size of the supervisory board, and the sum of the compensation of the top three executives. Hypothesis requires $\hat{\beta}_2 < 0$.

To examine the moderating role of financial constraint (SA_{it}), this paper augments model (1) by including the interaction of SA_{it} with the CIO variable and creates the following model:

$$PEG_{i,t} = \beta_0 + \beta_1 CIO_{i,t} + \beta_2 CIO_{i,t} \times SA_{i,t} + \delta \mathbf{X}_{it} + \varphi_i + \lambda_t + \varepsilon_{it} \quad (4)$$

where SA_{it} represents the level of financial constraint, which is constructed by the sensitivity of a firm's investment to changes in its cash flow based on the method proposed by Hadlock and Pierce (2010). Hypothesis requires $\hat{\beta}_2 < 0$.

4. Empirical results on the effect of CIO on COE

Table 1 presents the definition and measurement of the main variables. Table 2 presents the descriptive statistics. The mean COE is 10.7% with a standard deviation of 3.85%. The minimum is 2.93% and the maximum is 22.8%, suggesting that there is a large variation in COE among PLCs. In addition, the mean of CIO_1, is 0.103 with a standard deviation of 0.305. This indicates that the proportion of CIIs holding at least a 5% stake in two or more companies in the same industry is only 10.3%. The mean of CIO connectivity, CIO_2, is 1.080, with a maximum value of 4. This indicates that companies have a maximum of four common shareholders with shared ownership.

Table 3 presents the estimation results of the panel data model (1). Columns (i) and (ii) contain results without control variables or fixed effects, columns (iii) and (iv) contain results with control variables but without any fixed effects, while columns (v) and (vi) contain results with control variables as well as firm- and year-fixed effects. As shown in columns (i), (iii) and (v), the coefficient estimates for CIO_1_{it} are −0.008, −0.006 and −0.008, respectively. These are significant at the 1% level, suggesting that the presence of CIO leads to a 0.6 to 0.8 percentage point decrease in COE.

In addition, as shown in columns (ii), (iv) and (vi), the coefficient estimates for CIO_2_{it} are −0.011, −0.008, and −0.011, respectively. These are significant at the 1% level, suggesting that the presence of an additional CII for the company leads to a 0.8 to 1.1 percentage points decrease in COE. Hence, the findings from the panel data regression suggest that the hypothesis cannot be rejected. By fostering a more stable and predictable business environment, CIIs help lower the perceived risks associated with investing in the company. Thus, the company's COE decreases, reflecting the lower return investors require to compensate for reduced risks.

The above results are consistent with theoretical predictions that the exit threat posed by CIIs and the synergistic governance effect they create can help reduce a company's COE (Boone & White, 2015; Edmans et al., 2019; Koch et al., 2021; Li & Liu, 2023). When companies perceive the risk of CIIs reducing their shareholdings, they are incentivized to improve their corporate governance practices. This proactive approach is driven by the desire to avoid the negative signals and potential market repercussions that

Table 1. Definition and measurement of variables.

Type of Variable	Symbol	Measurement
Dependent	PEG	The cost of equity capital for a company, calculated using the PEG (Price/Earnings to Growth) model.
Core Independent	CIO_1	If there are common institutional investors holding shares in the company in the current year, it takes value of 1; otherwise, it takes value of 0.
	CIO_2	The natural logarithm of the sum of the number of common institutional investors holding shares in a listed company and 1.
	CIO_3	The sum of the percentage of shares held by all common institutional investors
Control	Size	Size of a company, the natural logarithm of its total assets.
	Lev	Leverage: The ratio of total liabilities to total assets.
	ROA	Return on Assets: The ratio of operating profit for the period to total assets at the end of the period.
	ROE	Profitability: The ratio of net profit to the average shareholders' equity.
	Liquid	Liquidity: The ratio of current assets to current liabilities.
	BM	Book to Market: The ratio of the market value to the book value of equity.
	Board	Board Size: The natural logarithm of the number of members on a company's board of directors.
	Indep	Percentage of independent directors: The ratio of the number of independent directors to the total number of directors on the board.
	Shrholder	Ownership of the largest shareholder: The ratio of the number of shares held by the largest shareholder to the total number of shares outstanding.
	INST	Institutional investor ownership: The ratio of the total number of shares held by institutional investors to the total number of shares outstanding.
	Growth	Revenue growth rate: The ratio of the current year's revenue to the previous year's revenue minus 1.
Moderator	HHI	Herfindahl-Hirschman index: The sum of the squared ratios of the revenue of each company in the industry to the total revenue of the industry.
	CG	Level of corporate governance: An index for corporate governance quality using principal component analysis on eight variables.
	SA	Index for financing constraint

Note. The indicator of CIO is manually collected and compiled based on quarterly financial reports available from the China Securities Market and Accounting Research (CSMAR) database. Other variables are obtained from both the CSMAR and Wind database.

Table 2. Descriptive statistics.

Variable	No. of Observations	Mean	Standard Deviation	Minimum	Maximum
CIO_1	35,635	0.103	0.305	0	1
CIO_2	35,635	1.080	1.260	0	4
CIO_3	34023	0.025	0.09	0	0.535
PEG	21,411	0.107	0.0385	0.0293	0.228
MPEG	20474	0.126	0.051	0.033	0.309
HHI	34018	0.505	0.200	0	1
CG	32413	0.002	0.064	−0.107	0.206
KZ	34023	1.089	2.330	−6.640	6.165
Size	33,994	22.17	1.310	19.86	26.27
Lev	33,994	0.408	0.204	0.0514	0.883
ROA	33,994	0.0361	0.0628	−0.218	0.217
ROE	33,994	0.0551	0.125	−0.588	0.351
Liquid	33,901	2.738	2.915	0.338	18.52
BM	33,901	0.626	0.258	0	1.182
Board	33,901	2.116	0.199	1.609	2.639
Indep	33,901	37.51	5.348	28.57	57.14
Shrholder	33,901	34.27	14.85	8.413	74.02
INST	33,901	44.31	25.38	0.204	92.95
Growth	34,022	0.146	0.366	−0.542	2.166

could arise from a reduction in CII ownership. Enhanced corporate governance, in turn, reassures investors, leading to lower risk perceptions and a subsequent decrease in COE.

Moreover, CIIs often bring valuable industry-specific knowledge, information, and experience. Their involvement can streamline monitoring processes and reduce associated costs by providing more efficient oversight. Additionally, CIIs can play a pivotal role in aligning and coordinating industry product market strategies, competition strategies, and dispute resolutions among companies in which they hold shares. This coordination not only helps in mitigating cutthroat competition but also stabilizes the market environment, reducing stock price volatility and the uncertainty faced by investors, which lowers a company's COE.

Regarding control variables, the regression results with firm- and year-fixed effects reveal the following findings: Firstly, firm size and leverage have significantly positive coefficients at the 1% level,

Table 3. Estimation Results on the Impact of Common Institutional Investor on Cost of Equity.

	(i) PEG	(ii) PEG	(iii) PEG	(iv) PEG	(v) PEG	(vi) PEG
CIO_1	−0.008***		−0.006***		−0.008***	
	(−7.06)		(−7.13)		(−6.89)	
CIO_2		−0.011***		−0.008***		−0.011***
		(−7.18)		(-7.04)		(-7.01)
Size			0.000*	0.000*	0.003***	0.003***
			(1.66)	(1.66)	(7.11)	(7.10)
Lev			0.029***	0.029***	0.007***	0.007***
			(14.68)	(14.69)	(2.70)	(2.70)
ROA			−0.085***	−0.085***	−0.024*	−0.024*
			(−6.71)	(−6.69)	(−1.77)	(−1.73)
ROE			0.038***	0.037***	0.001	0.001
			(5.59)	(5.57)	(0.16)	(0.13)
Liquid			0.001***	0.001***	0.000	0.000
			(6.86)	(6.86)	(1.10)	(1.11)
BM			0.010***	0.010***	−0.009***	−0.009***
			(7.87)	(7.90)	(−5.80)	(−5.77)
Board			−0.006***	−0.006***	−0.002	−0.003
			(−3.62)	(−3.56)	(−1.14)	(−1.15)
Indep			0.000	0.000	0.000	0.000
			(0.23)	(0.25)	(0.92)	(0.90)
Shrholder			−0.000	−0.000	−0.000	−0.000
			(−1.57)	(−1.59)	(−0.07)	(−0.07)
INST			−0.000***	−0.000***	−0.000*	−0.000*
			(−5.17)	(−5.15)	(−1.82)	(−1.84)
Growth			0.005***	0.005***	0.003***	0.003***
			(6.47)	(6.47)	(4.20)	(4.23)
Constant	0.120***	0.120***	0.093***	0.093***	0.053***	0.053***
	(145.60)	(145.54)	(13.47)	(13.41)	(4.53)	(4.55)
Year FE	N	N	N	N	Y	Y
Ind FE	N	N	N	N	Y	Y
Observations	21,411	21,411	20,584	20,584	20,584	20,584
Adj R2	0.156	0.156	0.046	0.046	0.166	0.166

Note. This table presents the results on the impact of common institutional ownership on the cost of equity capital. The dependent variable is constructed using the PEG model. Columns (i) and (ii) contain results without control variables or fixed effects. Columns (iii) and (iv) contain results with control variables but without any fixed effects. Columns (v) and (vi) contain results with control variables as well as firm- and year-fixed effects. Variable definitions are presented in Table 1. Cluster-robust standard errors at the firm level are reported in parentheses. ***, **, and *indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

indicating that larger firms and those with higher leverage tend to have higher COE. Secondly, growth opportunities also have a significantly positive coefficient at the 1% level, suggesting that firms with greater growth prospects experience higher COE. Thirdly, ROA has a significantly negative coefficient at the 10% level, indicating that firms with better performance tend to have lower COE. Fourthly, the book-to-market ratio has a significantly negative coefficient at the 1% level, suggesting that undervalued firms have lower COE. Lastly, regarding the effect of institutional ownership on COE, the results indicate a significant negative relationship. However, the estimated coefficients are close to zero, suggesting that the economic impact of institutional ownership on COE is negligible or minimal.

5. Robustness tests

5.1. Alternative measures of COE and CIO

In order to mitigate the potential impact of the measurement method for COE and CIO on the empirical results, this study followed the methodology proposed by Easton (2004) and employed the MPEG model to measure COE and assess the robustness of the regression results presented in regression (1). The updated results can be found in columns (i) and (ii) of Table 4.

The regression results reveal that the coefficient estimates for CIO_1_{it} and CIO_2_{it} are −0.010 and −0.014, respectively, and both are statistically significant at the 1% level. This suggests that when there is CIO within a PLC, on average, the COE of the company decreases by 1 percentage point. Moreover, for each additional CII in a PLC, the COE decreases by 1.4 percentage points. These findings align with the results reported in Table 4, further confirming the robustness of the baseline regression in this study.

Table 4. Alternative measure of common institutional investor and COE.

	(i) MPEG	(ii) MPEG	(iii) MPEG	(iv) PEG
CIO_1	−0.010***			
	(−6.84)			
CIO_2		−0.014***		
		(−6.94)		
CIO_3			−0.028***	−0.022***
			(-4.16)	(-4.16)
ROA	−0.040**	−0.039**	−0.039**	−0.023*
	(−2.21)	(−2.17)	(−2.15)	(−1.71)
ROE	0.005	0.005	0.005	0.001
	(0.55)	(0.51)	(0.54)	(0.16)
Liquid	−0.000	−0.000	−0.000	0.000
	(−1.20)	(−1.20)	(−1.17)	(1.14)
BM	−0.014***	−0.014***	−0.014***	−0.009***
	(−7.06)	(−7.04)	(−7.21)	(-5.93)
Board	−0.003	−0.003	−0.003	−0.002
	(−1.03)	(−1.04)	(−0.97)	(−1.08)
Indep	0.000	0.000	0.000	0.000
	(0.45)	(0.43)	(0.50)	(0.96)
Shrholder	−0.000	−0.000	−0.000	−0.000
	(−1.47)	(−1.47)	(−1.55)	(−0.14)
INST	−0.000*	−0.000*	−0.000*	−0.000*
	(−1.75)	(−1.77)	(−1.67)	(−1.75)
Growth	0.002**	0.002**	0.002**	0.003***
	(2.25)	(2.27)	(2.24)	(4.19)
	(9.55)	(9.56)	(9.03)	(5.79)
Constant	0.073***	0.074***	0.074***	0.053***
	(4.86)	(4.89)	(4.90)	(4.56)
Year FE	Y	Y	Y	Y
Ind FE	Y	Y	Y	Y
Observations	20,184	20,184	20,184	20,584
Adj R2	0.190	0.191	0.189	0.165

Note. This table presents the results on the impact of common institutional ownership on the cost of equity capital using alternative measures for COE and CIO. Cluster-robust standard errors at the firm level are reported in parentheses. ***, **, and *indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 5. Endogeneity test.

	First stage		Second stage	
	(i) CIO_1	(ii) CIO_2	(iii) PEG	(iv) PEG
L.CIO_1	0.851***			
	(105.87)			
L.CIO_2		0.859***		
		(111.13)		
CIO_1			−0.00566**	
			(-3.86)	
CIO_2				−0.00721**
				(-3.77)
F-stat (First Stage)	1361.91***	1334.19***		
Control	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Ind FE	Y	Y	Y	Y
Observations	17576	17576	17576	17576

Note. This table presents the Two-stage Least Squares (2SLS) estimation results to examine the possibility of reverse causality. Lagged CIO is used as the instrumental variable. Columns (i) and (ii) contain the results from the first stage regressions. columns (iii) and (iv) report the results from the second-stage regressions. Cluster-robust standard errors at the firm level are reported in parentheses. ***, **, and *indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Furthermore, in line with He and Huang (2017) and Chen et al. (2024), this study employs an alternative measure, denoted as CIO_3_{it}, to capture the level of CIO. This measure represents the cumulative ownership stakes of all CIIs in a PLC. The updated regression results are presented in columns (iii) and (iv) of Table 5. Based on the regression results, the coefficient estimates for CIO_3_{it} are −0.028 and −0.022 when the COE is calculated using the MPEG model and the PEG model, respectively. Both coefficient estimates are statistically significant at the 1% level. These findings indicate that even after modifying the measurement method for CIO, the impact of CIO on the COE remains robust.

Table 6. The Moderating role of industry competition, corporate governance and financial constraint.

	Industry Competition		Corporate Governance		Financial Constraint	
	(i)	(ii)	(iii)	(iv)	(v)	(vi)
CIO_1*HHI	0.522***					
	(3.75)					
CIO_2*HHI		0.527***				
		(3.64)				
CIO_1*CG			−0.014***			
			(−6.32)			
CIO_2*CG				−0.019***		
				(-6.43)		
CIO_1*SA					−0.304***	
					(-3.32)	
CIO_2*SA						−0.421***
						(-3.33)
CIO_1	−0.012***		−0.011***		−0.007***	
	(−5.69)		(−8.80)		(−5.83)	
CIO_2		−0.011***		−0.009***		−0.006***
		(−6.02)		(-8.26)		(-5.53)
HHI	−0.000	−0.000				
	(−0.13)	(−0.16)				
CG			0.010	0.009		
			(1.53)	(1.47)		
SA					−0.002***	−0.002***
					(−8.45)	(−8.68)
Size	0.003***	0.003***	0.003***	0.003***	0.003***	0.003***
	(7.23)	(7.22)	(7.18)	(7.18)	(7.14)	(7.15)
Lev	0.007***	0.007***	0.007***	0.007***	0.007***	0.007***
	(2.76)	(2.77)	(2.72)	(2.70)	(2.73)	(2.72)
ROA	−0.022	−0.023*	−0.024*	−0.024*	−0.024*	−0.024*
	(−1.64)	(−1.65)	(−1.76)	(−1.72)	(−1.75)	(−1.74)
ROE	0.000	0.001	0.001	0.001	0.001	0.001
	(0.07)	(0.08)	(0.17)	(0.13)	(0.18)	(0.17)
Liquid	0.000	0.000	0.000	0.000	0.000	0.000
	(1.15)	(1.17)	(1.08)	(1.08)	(1.10)	(1.12)
BM	−0.009***	−0.009***	−0.009***	−0.009***	−0.009***	−0.009***
	(−6.00)	(−5.98)	(−5.87)	(−5.85)	(−5.95)	(−5.93)
Board	−0.002	−0.002	−0.003	−0.003	−0.002	−0.002
	(−1.13)	(−1.13)	(−1.16)	(−1.16)	(−1.11)	(−1.11)
Indep	0.000	0.000	0.000	0.000	0.000	0.000
	(1.00)	(0.97)	(0.95)	(0.92)	(0.98)	(0.98)
Shrholder	−0.000	−0.000	0.000	0.000	−0.000	−0.000
	(−0.11)	(−0.10)	(0.03)	(0.03)	(−0.07)	(−0.07)
INST	−0.000*	−0.000*	−0.000*	−0.000*	−0.000*	−0.000*
	(−1.71)	(−1.73)	(−1.76)	(−1.77)	(−1.73)	(−1.74)
Growth	0.003***	0.003***	0.003***	0.003***	0.003***	0.003***
	(4.11)	(4.12)	(4.16)	(4.19)	(4.09)	(4.09)
Constant	0.051***	0.051***	0.051***	0.052***	0.052***	0.052***
	(4.35)	(4.37)	(4.44)	(4.45)	(4.45)	(4.44)
Year FE	Y	Y	Y	Y	Y	Y
Ind FE	Y	Y	Y	Y	Y	Y
Adj R2	0.165	0.164	0.166	0.166	0.164	0.164

Note. This table presents the results on the moderation effect of common institutional ownership and the cost of equity capital. Columns (i) and (ii) contain results with industry competition as the moderator. Columns (iii) and (iv) contain results with corporate governance as the moderator. Columns (v) and (vi) contain results with financial constraints as the moderator. Cluster-robust standard errors at the firm level are reported in parentheses. ***, **, and *indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

5.2. Endogeneity tests

One potential issue in this study is the influence of firm-specific attributes, such as size, profitability, market share, and managerial characteristics, on the CIO. These factors can affect the investment choices of institutional investors, potentially leading to endogeneity problems. To address this concern, the study conducts the Heckman test to examine possible sample selection biases. The results of the Heckman test fail to reject the null hypothesis that the inverse Mill's ratio is equal to zero, indicating no statistically significant endogeneity issues resulting from sample selection biases in the regression results.

Additionally, the experience and informational advantages of common institutional investors themselves may contribute to their preference for investing in higher-value and lower-risk firms. This could potentially result in a significant reduction in the COE. Therefore, the findings presented in Tables 3 and

4 may not solely reflect the synergistic governance effects of CIO but rather the lower risk faced by the firms in which CIIs invest. To further address the possibility of reverse causality, the study employs lagged CIO as an instrumental variable and utilizes Two-stage Least Squares (2SLS) estimation. The estimation results of 2SLS are reported in Table 5.

The first-stage regression results are presented in columns (i) and (ii). These results indicate that the coefficient estimates of L.CIO_1 and L.CIO_2 are statistically significant and positive, suggesting a positive correlation between the instrumental variables and the core explanatory variables. The F-statistics of 1361.91 and 1334.19 indicate the absence of weak instrument problems, thus supporting the appropriateness of using lagged CIO as instrumental variables.

The second-stage regression results are reported in columns (iii) and (iv). These results demonstrate that, after accounting for potential endogeneity issues using instrumental variables, the impact of CIO on COE remains significantly negative at the 5% level. Therefore, the empirical findings of this study are robust to potential endogeneity problems.

6. Further analysis

6.1. Moderation effects

Regarding the moderating effect of industry competition, the results in Table 6, columns (i) and (ii), indicate that the coefficients for the interaction terms between CIO_1 and HHI ($CIO_1_{i,t} \times HHI_{i,t}$) and CIO_2 and HHI ($CIO_2_{i,t} \times HHI_{i,t}$) are 0.522 and 0.527, respectively. Both coefficients are statistically significant at the 1% level. Specifically, a 10-percentage point decrease in HHI (indicating increased industry competition) results in an additional decrease in the COE of 5.22 to 5.27 percentage points. Therefore, the hypothesis cannot be rejected.

Regarding the moderating effect of corporate governance, the results in Table 6, columns (iii) and (iv), show that the coefficient estimates for the interaction terms between CIO_1 and the corporate governance index ($CIO_1_{i,t} \times CG_{i,t}$) and CIO_2 and the corporate governance index ($CIO_2_{i,t} \times CG_{i,t}$) are −0.014 and −0.019, respectively. Both coefficients are statistically significant at the 1% level. Specifically, a one-point increase in the corporate governance index leads to an additional decrease in the COE of 1.4 to 1.9 percentage points. This outcome is consistent with the hypothesis.

Regarding the moderating effect of financial constraint, the results presented in Table 6, columns (v) and (vi), reveal that the coefficient estimates for the interaction terms between CIO_1 and the SA index ($CIO_1_{i,t} \times SA_{i,t}$) and CIO_2 and the SA index ($CIO_2_{i,t} \times SA_{i,t}$) are −0.304 and −0.421, respectively. Both coefficients are statistically significant at the 1% level. Specifically, a one-percentage-point increase in the SA index leads to an additional decrease in the COE of 0.304 to 0.421 percentage points. The results confirm that as the level of financial constraint increases, the governance effect exerted by CIO becomes more significant. Thus, hypothesis cannot be rejected.

6.2. Heterogeneity analysis

It is well documented that there are significant differences in corporate governance and operational objectives between state-owned enterprises (SOEs) and non-SOEs. SOEs often have political tasks in their decision-making processes, which weakens the supervision by institutional investors and diminishes the synergistic governance effect. In contrast, non-SOEs usually prioritize profit objectives and have a stronger motivation to engage in active corporate governance, aligning more closely with CIIs. Consequently, the impact of CIIs in facilitating corporate governance is more pronounced in non-SOEs, leading to a larger reduction in the COE.

To further investigate this, the study divides the sample into two subsamples based on whether a PLC is an SOE and re-estimates regression (1). The empirical results are presented in Table 7. For the sample of SOEs, the coefficient estimates for CIIs are −0.004 and −0.005, both statistically significant at the 5% level. In the sample of non-SOEs, the coefficient estimates for CIIs are −0.011 and −0.015, both statistically significant at the 1% level. The absolute value of the coefficient for non-SOEs is greater than

Table 7. Heterogeneity analysis based on ownership structure.

	State-owned Enterprises		Non-state-owned Enterprises	
	(i)	(ii)	(iii)	(iv)
CIO_1	−0.004**		−0.011***	
	(−2.04)		(−7.32)	
CIO_2		−0.005**		−0.015***
		(−2.22)		(−7.23)
Size	0.003***	0.003***	0.003***	0.003***
	(3.44)	(3.43)	(4.36)	(4.37)
Lev	−0.001	−0.001	0.010***	0.010***
	(−0.22)	(−0.21)	(3.19)	(3.19)
ROA	−0.059**	−0.059**	0.003	0.004
	(−2.14)	(−2.14)	(0.20)	(0.25)
ROE	0.024**	0.024**	−0.013	−0.014
	(1.97)	(1.97)	(−1.56)	(−1.61)
Liquid	0.001*	0.001*	0.000	0.000
	(1.83)	(1.82)	(0.15)	(0.18)
BM	−0.008***	−0.008***	−0.009***	−0.009***
	(−2.87)	(−2.86)	(−4.70)	(−4.66)
Board	0.002	0.002	−0.004	−0.004
	(0.47)	(0.47)	(−1.52)	(−1.54)
Indep	0.000	0.000	0.000	0.000
	(0.18)	(0.18)	(0.86)	(0.83)
Shrholder	−0.000	−0.000	0.000	0.000
	(−0.40)	(−0.41)	(0.74)	(0.76)
INST	−0.000	−0.000	−0.000*	−0.000*
	(−0.58)	(−0.59)	(−1.95)	(−1.95)
Growth	0.001	0.001	0.004***	0.004***
	(0.69)	(0.71)	(4.68)	(4.69)
	(0.86)	(0.87)	(7.26)	(7.24)
Constant	0.050**	0.050**	0.071***	0.071***
	(2.32)	(2.33)	(4.81)	(4.81)
Year FE	Y	Y	Y	Y
Ind FE	Y	Y	Y	Y
Observations	6,344	6,344	14,240	14,240
Adj R2	0.128	0.129	0.196	0.195
F-test			123.0***	122.9***

Note. This table presents the results on the heterogeneous impact of ownership structure on the relationship between common institutional ownership and the cost of equity capital. Cluster-robust standard errors at the firm level are reported in parentheses. ***, **, and *indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

that for SOEs. The F-statistics are 123.0 and 122.9 with a *p*-value of 0.000, indicating that the null hypothesis of no difference in the coefficient estimates for the SOE and non-SOE subsamples can be rejected at the 1% level. These findings suggest that SOEs face challenges in fully harnessing the benefits of CIIs' supervision and governance role. Conversely, non-SOEs experience greater pressure from the potential withdrawal of CIIs. Therefore, it can be concluded that the effect of CIIs in reducing the COE is more significant for non-SOEs.

Furthermore, considering the variations in regional economic development levels, the study further divides the sample into the eastern and non-eastern regions to examine regional heterogeneity. Table 8 presents the results of this analysis. Columns (i) and (ii) represent the non-eastern region, with coefficient estimates of − 0.006 and −0.009, respectively. Columns (iii) and (iv) represent the eastern region, with coefficients of −0.009 and −0.012, respectively. All of these coefficients are statistically significant at the 1% level. The F-statistics are 78.82 and 78.84 with a *p*-value of 0.000, indicating that the null hypothesis of no difference in the coefficient estimates for the eastern and non-eastern regions can be rejected at the 1% level.

These findings suggest that the impact of CIO on the COE is weaker in the non-eastern region compared to the eastern region. One potential explanation is that the eastern region, characterized by higher levels of economic development and more intense industry competition, has companies that are more sensitive to the threat of withdrawal by CIIs and more likely to benefit from the synergistic governance effect. Conversely, the non-eastern region, with a smaller observed sample of companies and relatively lower levels of development and industry competition, exhibits a weaker impact of CIO on the reduction of COE.

Table 8. Heterogeneity Analysis Based on Regional Differences.

	Non-eastern region		Eastern region	
	(i)	(ii)	(iii)	(iv)
CIO_1	−0.006***		−0.009***	
	(−2.89)		(−5.20)	
CIO_2		−0.009***		−0.012***
		(−2.89)		(−5.24)
Size	0.003***	0.003***	0.004***	0.004***
	(3.26)	(3.24)	(5.31)	(5.32)
Lev	0.005	0.005	0.007*	0.007*
	(0.93)	(0.91)	(1.84)	(1.86)
ROA	−0.038	−0.037	−0.050**	−0.049**
	(−1.42)	(−1.39)	(-2.35)	(-2.35)
ROE	0.010	0.009	0.017	0.017
	(0.77)	(0.74)	(1.55)	(1.55)
Liquid	0.000	0.000	−0.000	−0.000
	(1.60)	(1.61)	(-0.37)	(-0.37)
BM	−0.007**	−0.007**	−0.010***	−0.010***
	(−2.46)	(−2.43)	(−4.62)	(−4.60)
Board	0.002	0.002	−0.004	−0.005
	(0.53)	(0.54)	(−1.37)	(−1.40)
Indep	0.000	0.000	0.000	0.000
	(0.37)	(0.37)	(0.25)	(0.21)
shrholder1	−0.000	−0.000	−0.000	−0.000
	(−0.46)	(−0.47)	(−0.45)	(−0.45)
INST	−0.000	−0.000	−0.000**	−0.000**
	(−0.96)	(−0.96)	(−2.17)	(−2.19)
Growth	0.002	0.002	0.004***	0.004***
	(1.34)	(1.36)	(3.77)	(3.79)
Constant	0.047**	0.047**	0.056***	0.056***
	(2.00)	(2.01)	(3.32)	(3.34)
Year FE	Y	Y	Y	Y
Ind FE	Y	Y	Y	Y
Observations	5,568	5,568	10,143	10,143
Adj R2	0.137	0.137	0.181	0.181
F-test			78.82***	78.84***

Note. This table presents the results on the heterogeneous impact of eastern versus non-eastern regions on the relationship between common institutional ownership and the cost of equity capital. Cluster-robust standard errors at the firm level are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

7. Conclusions and implications

This paper presents an empirical analysis of the impact of CIO on the COE, using all non-financial companies publicly listed in China during 2012 and 2022. The study finds that CIO significantly reduces the COE for PLCs, aligning with traditional asset pricing theories that suggest CIO is associated with a reduction in a firm's non-diversifiable risk. This risk mitigation may result from the exit threat posed by CIIs and their synergistic governance effects. When companies perceive the potential negative signals and market consequences from CIIs reducing their shareholdings, they are incentivized to improve their corporate governance practices. This enhancement reduces investor risk expectations, leading to a lower COE. Furthermore, CIIs contribute industry-specific knowledge, information, and expertise, which helps lower monitoring costs and uncertainty. They can also facilitate the coordination of industry strategies, such as product market strategies, competitive approaches, and dispute resolution, which collectively contribute to reduced stock price volatility and investor risk expectations. Consequently, these factors lead to a decreased COE for the company.

The study also finds that the impact of CIO on the COE is stronger for firms operating in more competitive industries, those with better corporate governance, and those facing higher financing constraints. Furthermore, the effect of CIO on the COE is more pronounced for non-SOEs and companies located in economically developed eastern regions.

The implications of these findings can be categorized into three main aspects. Firstly, the study emphasizes the importance of fully harnessing the advantages of CIO in corporate supervision, management, and synergistic governance. Rather than focusing on short-term gains from collusive misconduct, CIIs should utilize their information resources and management expertise to promote cooperative efforts, drive innovation and development, and enhance governance models. These actions can reduce unethical

competition, lower information costs and the COE, and help companies establish long-term business advantages.

Secondly, given that CIO may lead to collusive behavior, monopolistic competition, and inefficiency in resource allocation in capital markets, relevant authorities should strengthen the supervision and regulation of CIO. This may involve revising and implementing anti-monopoly laws and regulations, maintaining a fair market environment, preventing disorderly capital expansion, and remaining vigilant about potential collusive monopolistic practices arising from CIO.

Thirdly, relevant authorities should actively establish mechanisms to mitigate the inefficiencies caused by regional market segmentation. They can leverage the synergistic governance role of CIO, reduce financing barriers, improve financial service capabilities, and create a favorable financial ecosystem that benefits companies and investors alike.

While this study yields interesting results and implications, it is important to acknowledge several limitations. Firstly, the study primarily focuses on CIO indicators constructed using data from PLCs, which may overlook the potential influence of institutional investors simultaneously holding shares in non-listed companies. This limitation restricts the analysis to a subset of the overall CIO landscape. Future studies could enhance the robustness and comprehensiveness of CIO measures by incorporating data on institutional investors' holdings in both listed and non-listed companies.

Secondly, the measurement of the COE itself introduces certain challenges and complexities. While this study employs PEG and MPEG models to measure COE, it is important to note that there is no consensus on the best approach. Future research could explore more elaborate measures of the COE, taking into account additional factors such as firm-specific risk, market conditions, and investor expectations.

Thirdly, the study relies on data from non-financial Chinese PLCs. This may restrict the generalizability of the findings to other countries. The dynamics between CIO and the COE may vary across different market conditions, regulatory environments, and industry contexts. Future research could include a broader range of countries or regions to provide a more comprehensive understanding of the relationship between CIO and the COE in different contexts. This would enhance the external validity of the findings and provide a more robust foundation for drawing broader conclusions.

Note

1. The China Securities Market and Accounting Research (CSMAR) database is widely regarded as a reliable and comprehensive source of financial and economic data for Chinese markets. The database is known for its high standards of data accuracy. It employs rigorous quality control measures, including multiple layers of data verification and validation, to ensure the reliability of the information provided. CSMAR is regularly updated to reflect the latest available data, making it a vital resource for those studying Chinese financial markets. Extensively utilized by academic researchers, financial analysts, and policymakers, CSMAR has become a cornerstone for empirical research in the fields of finance, accounting, and economics. Many articles published in top economic and finance journals have used CSMAR data, underscoring its importance and trustworthiness in the academic and professional communities.

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Larry Su hold a PhD in Economics and have taught at University of Greenwich, Coventry University, University Technology Brunei prior to joining in Edge Hill Business School. Much of his current research has focused on corporate internationalization strategies, managerial compensation, geographical structure of financial institutions, and green finance.

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Data availability statement

Data will be available upon request.

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